

Chapter Two

Binding Energy

It has been seen that the predicted mass $(Zm_p + (A - Z)m_n)$ is more than the actual mass M .

This additional mass is accounted for by binding energy that is released when a nucleus is formed.

The difference in predicted mass and actual mass is identified as mass defect ΔM .

For any nucleus composition ${}_Z^AX$ mass defect is $\Delta M = [Zm_p + (A - Z)m_n] - M$

If we want to break the nucleus of ${}_Z^AX$ into Z number of proton and $A - Z$ number of neutrons we need to supply ΔMc^2 amount of energy to nucleus. The energy equivalent to mass defect is identified as Binding energy $E_b = \Delta mc^2$

The convenient measure to analyse Binding energy of nucleus ratio between binding energy E_b and its constituents i.e. A . If we plot binding energy per nucleon $\frac{E_b}{A}$ as number of mass number we get following curve.

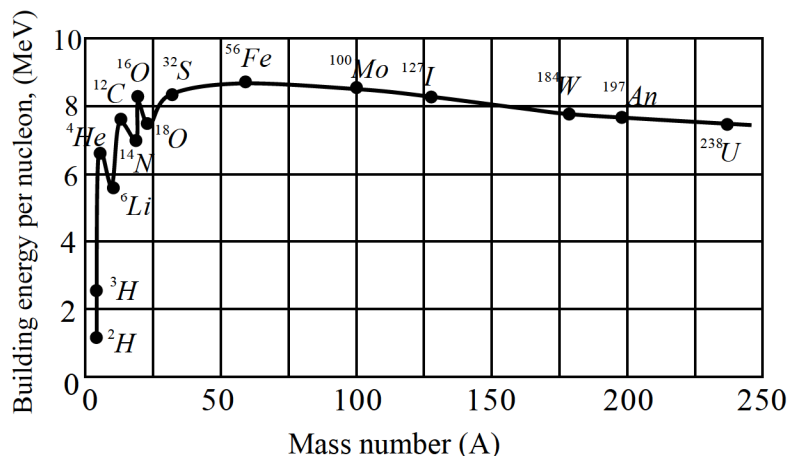


Figure: The binding energy per nucleon as a function of mass number

Main feature of Binding energy per nucleon versus the mass number

(a) The binding energy per nucleon, $\frac{E_b}{A}$, is practically constant, i.e. practically independent of the atomic number for nuclei of middle mass number ($30 < A < 170$). The curve has a maximum of about 8.75 MeV for $A = 56$ MeV and has a value of 7.6 MeV for $A = 238$

(b) $\frac{E_b}{A}$ is lower for both light nuclei $A < 30$ and $A > 170$

We can draw some conclusions from these two observations:

- (i) The force is attractive and sufficiently strong to produce a binding energy of a few MeV per nucleon.
- (ii) The constancy of the binding energy in the range $30 < A < 170$ is a consequence of the fact that the nuclear force is short-ranged.

Consider a particular nucleon inside a sufficiently large nucleus. It will be under the influence of only some of its neighbours, which come within the range of the nuclear force. If any other nucleon is at a distance more than the range of the nuclear force from the particular nucleon it will have no influence on the binding energy of the nucleon under consideration.

(iii) A very heavy nucleus, say $A = 240$, has lower binding energy per nucleon compared to that of a nucleus with $A = 120$. Thus, if a nucleus $A = 240$ breaks into two $A = 120$ nuclei, nucleons get more tightly bound. This implies energy would be released in the process. It has very important implications for energy production through fission,

(iv) Consider two very light nuclei ($A \leq 10$) joining to form a heavier nucleus. The binding energy per nucleon of the fused heavier nuclei is more than the binding energy per nucleon of

the lighter nuclei. This means that the final system is more tightly bound than the initial system.

Again, energy would be released in such a process of fusion

Example: If mass of ${}^4_2\text{He}$ nucleus is 4.002602u find the binding energy of ${}^4_2\text{He}$. Also calculate

Binding energy per nucleon for ${}^4_2\text{He}$.

Solution: $E_b = \Delta Mc^2 = \left([Zm_p - (A - Z)m_n] - M \right) c^2$

$$E_b = \left[(2 \times 1.0078u + (4 - 2) \times 1.00866u) - 4.002602u \right] c^2 = 0.030378uc^2$$

$$E_b = 0.030378uc^2 \times \frac{931\text{MeV}}{c^2} = 28.3\text{MeV}$$

$$\frac{B.E}{A} = \frac{28.3}{4} = 7.07\text{MeV / nucleon}$$

Example: The measured mass of deuteron atom (${}^2_1\text{H}$), Hydrogen atom (${}^1_1\text{H}$), proton and neutron is 2.01649 u, 1.00782 u, 1.00727 u and 1.00866 u. Then find the binding energy of the deuteron nucleus (unit MeV/nucleon).

Solution: B.E = $[Z M_H + N M_N - M({}^2_1\text{H})] \times 931.5$

$$= [1 \times 1.00782 + 1 \times 1.00866 - 2.01649] \times 931.5$$

$$= (0.00238) \times 931.5 = 2.224 \text{ MeV}$$

Example: The binding energy of the neon isotope ${}^{20}_{10}\text{Ne}$ is 160.647 MeV. Find its atomic mass.

Solution: Here $Z=10$ and $A=10$, $M({}^A_Z\text{X}) = [ZM_H + NM_N] - \frac{B}{931.5 \text{ MeV / u}}$

$$M({}^{20}_{10}\text{Ne}) = \left[10(1.00782) + 10(1.00866) \right] - \frac{160.647}{931.5 \text{ MeV / u}} = 19.992 \text{ u}$$

Example:

(a) Find the energy needed to remove a neutron from the nucleus of the calcium isotope ${}^{42}_{20}\text{Ca}$.

(b) Find the energy needed to remove a proton from this nucleus.

(c) Why are these energies different?

Given: atomic masses of ${}^{42}_{20}\text{Ca} = 41.958622u$, ${}^{41}_{20}\text{Ca} = 40.962278u$, ${}^{41}_{19}\text{K} = 40.961825u$,

and mass of ${}^1_0n = 1.008665u$, ${}^1_1p = 1.007276u$.

Solution:

(a) ${}_{20}^{42}\text{Ca} \rightarrow {}_{20}^{41}\text{Ca} + {}_0^1n$; Total mass of the ${}_{20}^{41}\text{Ca}$ and ${}_0^1n = 41.970943 \text{ u}$.

Mass defect $\Delta m = 41.970943 - 41.958622 = 0.012321 \text{ u}$

So, B.E. of missing neutron $= \Delta m \times 931.5 = 11.48 \text{ MeV}$.

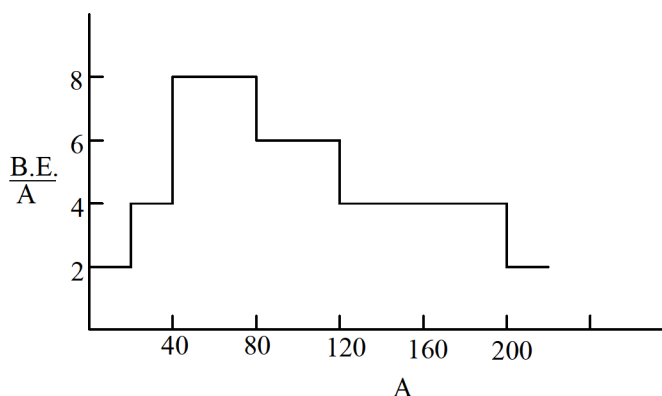
(b) ${}_{20}^{42}\text{Ca} \rightarrow {}_{19}^{41}\text{K} + {}_1^1p$; Total mass of the ${}_{19}^{41}\text{K}$ and ${}_1^1p = 41.919101 \text{ u}$.

Mass defect $\Delta m = 41.919101 - 41.958622 = 0.010479 \text{ u}$

So, B.E. of missing neutron $= \Delta m \times 931.5 = 10.27 \text{ MeV}$.

(c) The neutron was acted upon only by attractive nuclear forces whereas the proton was also acted upon by repulsive electric forces that decrease its binding energy.

Example: The following histogram represents the binding energy per particle ($B.E./A$) in MeV as a function of the mass number A of a nucleus. A nucleus with mass number $A = 180$ fissions into two nuclei of equal masses.



(a) Find the Binding energy for parents' nucleus $A = 180$

(b) Find the Binding energy of product nucleus (sum of binding energy of individual nucleus)

(c) how much energy is released /absorbed during fission

Solution: $A \rightarrow \frac{A}{2} + \frac{A}{2}$ or $180 \rightarrow 90 + 90$

(a) B.E. of nucleus $A = 180 \times 4 = 720 \text{ MeV}$

(b) Binding energy of Product $= 90 \times 6 + 90 \times 6 = 1080 \text{ MeV}$

(c) Since, B.E of the product nucleus is greater than the nucleus A , hence in this process energy is released and that is $= (1080 - 720) \text{ MeV} = 360 \text{ MeV}$.

Nuclear Force:

The binding energy per nucleon is approximately 8MeV which is much larger than binding energy of atom. There are various experiments carried out during 1930 to 1950 which give the following conclusion about nuclear forces.

- (1) The nuclear force is much stronger than the Coulomb force acting between charges or the gravitational forces between masses.
- (2) The nuclear force between two nucleons falls rapidly to zero as their distance is more than a few femtometres. This leads to saturation of forces in a medium or a large-sized nucleus, which is the reason for the constancy of the binding energy per nucleon.

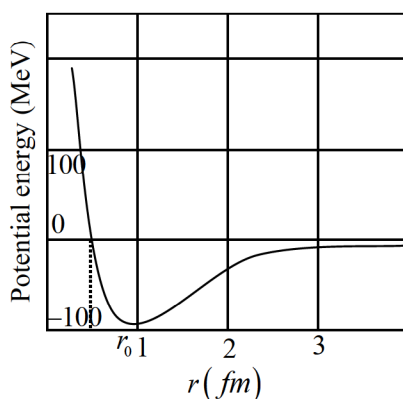


Figure: Potential energy of a pair of nucleons as a function of their separation. For a separation greater than r_0 , the force is attractive and for separations less than r_0 , the force is strongly repulsive.

The best approximation of nuclear potential is Yukawa potential given by Japanese scientist Hideki Yukawa. The Yukawa potential is given by $V(r) = g^2 \frac{\exp(-\alpha mr)}{r}$ where m is mass of

particle g^2 and α is positive constant. For $m = 0$, $g^2 = \frac{q_1 q_2}{4\pi\epsilon_0}$ Yukawa potential will approach to

Coulomb potential

A rough plot of the potential energy between two nucleons

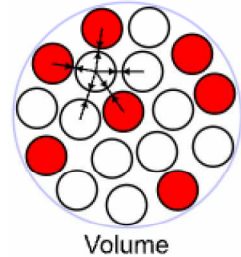
Liquid Drop Model of Binding Energy

This model is given by George Gamow, but finally it is proposed by Carl Friedrich Weizsacker. In theory we will see how nucleus behaves as drop of liquid. The liquid drop accounts for the spherical shape of most of nuclei and makes a rough prediction of binding energy. It is also known as semi-empirical mass formula.

We start assuming that energy associated with each nucleon-nucleon bond has some value U . This energy is negative since attractive forces are involved, but is usually written as positive because binding energy is considered a positive for convenience.

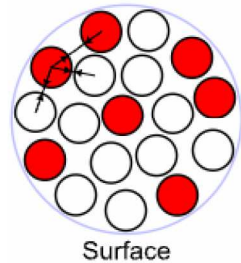
The corresponding mass formula is defined purely in term of number of protons and neutrons it contains. the original Weizsacker formula defines five terms

(1) **Volume Energy:** When an assembly of nucleons of the same size is packed together into the smallest volume, each interior nucleon has a certain number of other nucleons in contact with it. So, this nuclear energy is proportional to the volume. each bound energy is shared by two nucleons and twelve other spherical nucleons are in contact with is so each interior nucleons has binding energy $12 \times \left(\frac{1}{2}\right)U = 6U$.



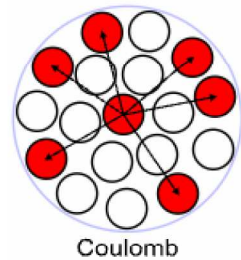
If all nucleons A in a nucleus were in the interior, the total binding energy would be $6AU$ the volume energy of nucleus can be written as $E_V = a_V A$.

(2) **Surface Energy:** Nucleons on the surface of every of every nucleus and therefore have fewer than twelve neighbours. the number of these nucleons depends on surface area of nucleus. The surface area is $4\pi R^2 = 4\pi R_0^2 A^{2/3}$. Hence the number of nucleons with fewer than the maximum number of bonds is proportional to $A^{2/3}$.



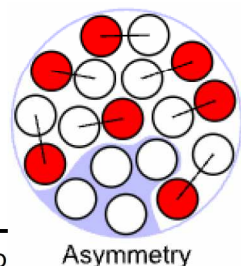
So, surface energy is $E_S = -a_S A^{2/3}$.

(3) **Coulomb Energy:** This term arises due to electrical repulsion between proton and proton. The coulomb energy of nucleus is a work that must be done to bring Z electron from infinity so average coulomb energy is given by $\frac{Z(Z-1)}{2} \cdot \left(-\frac{e^2}{4\pi\epsilon_0} \right) \left\langle \frac{1}{r} \right\rangle$ where $r \propto A^{1/3}$. So coulomb energy is given by



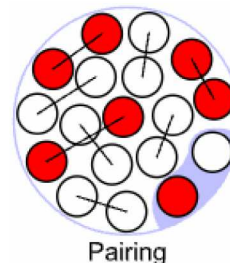
$$-a_C \frac{Z(Z-1)}{A^{1/3}}.$$

(4) **Asymmetry Energy:** (also called Pauli Energy), which accounts for the (Pauli exclusion principle). Unequal numbers of neutrons and protons imply filling higher energy levels for one type of particle, while leaving lower energy levels vacant for the other type.



This is a quantum effect arising from the Pauli exclusion principle which only allows two protons or two neutrons (with opposite spin direction) in each energy state. If a nucleus contains the same number of protons and neutrons then for each type the protons and neutrons fill to the same maximum energy level (the 'fermi level'). If, on the other hand, we exchange one of the neutrons by a proton then that proton would be required by the exclusion principle to occupy a higher energy state, since all the ones below it are already occupied. The upshot of this is that nuclides with $Z = N = A - Z$ have a higher binding energy, whereas for nuclei with different numbers of protons and neutrons (for fixed A) the binding energy decreases as the square of the number difference. The spacing between energy levels is inversely proportional to the volume of the nucleus - this can be seen by treating the nucleus as a three-dimensional potential well- and therefore inversely

Pairing Term: It is found experimentally that two protons or two neutrons bind more strongly than one proton and one neutron. The pairing term is positive for even-even nuclei such as ${}^4_2\text{He}$, ${}^{12}_6\text{C}$, ${}^{16}_8\text{O}$. it is zero for even-odd nuclei and negative for odd-odd nuclei. this term seems to vary as $A^{-3/4}$



So binding energy according to Weizsacker is

$$E_B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} - (\pm, 0) \frac{a_p}{A^{3/4}}$$

The coefficient that give good data is

$$a_v = 14.1 \text{ MeV} \quad a_s = 13.0 \text{ MeV} \quad a_c = 0.595 \text{ MeV} \quad a_A = 19 \text{ MeV} \quad a_p = 33 \text{ MeV}$$

Last two term Asymmetric and pairing term is correction to liquid drop model finally known as Semi-empirical mass formula for Binding energy.

Stability of nucleus

Example: In the liquid drop model the binding term due to coulomb repulsion is $-a_c \frac{Z(Z-1)}{A^{1/3}}$,

find the value of a_c

Solution: This term represent electric repulsion of the proton is negative term in binding energy.

The charge density $\rho = \frac{Ze}{\frac{4\pi}{3} R^3}$

Consider a thin spherical shell of this charge of radius $r < R$ using coulomb's law we can calculate the work dW required to bring this shell up to its radius r

$$dW = \frac{1}{4\pi\epsilon_0 r} \left[\frac{4}{3} \pi r^3 \rho \times 4\pi r^2 dr \rho \right]$$

$$W = \frac{1}{4\pi\epsilon_0} \frac{(4\pi)^2}{3} \left(\frac{Ze}{\frac{4\pi}{3} R^3} \right)^2 \int_0^R r^4 dr = \frac{3}{5} \frac{Z^2 e^2}{4\pi\epsilon_0 R}$$

In the equation of binding energy $B = (Zm_p + Nm_n - M)c^2$ it is assume that there is not already Z number of proton existed in the nucleus .

When we have only one proton in nucleus the $(B_c)_{proton} = \frac{3}{5} \frac{e^2}{4\pi\epsilon_0 R}$

For nucleus only one proton No work is done against the coulomb repulsion in assembling the nucleus.

In calculation of work done $W = \frac{3}{5} \frac{Z^2 e^2}{4\pi\epsilon_0 R}$ those term is also there which is binding energy of

single protons .so we must subtract Z time such a term

So binding energy due to coulomb repulsion is

$$\begin{aligned} -B_c &= \left(\frac{3Z^2 e^2}{5(4\pi\epsilon_0)R} - \frac{3}{5} \frac{Ze^2}{4\pi\epsilon_0 R} \right) = -\frac{3}{5} \frac{e^2 Z(Z-1)}{4\pi\epsilon_0 R} \\ \frac{3}{5} m_e c^2 \frac{e^2}{4\pi\epsilon_0 m_e c^2} \frac{Z(Z-1)}{1.2 \times A^{1/3}} \text{ where } m_e c^2 &= 0.511 \text{ MeV} \\ \frac{e^2}{4\pi\epsilon_0 m_e c^2} &= \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(9.11 \times 10^{-31})(3 \times 10^8)^2} = 2.8 \times 10^{-15} \text{ m} \end{aligned}$$

$$\text{So } -B_c = \frac{3}{5} \times 0.511 \times \frac{2.8 \times 10^{-15}}{1.2 \times 10^{-15}} = 0.66 \text{ MeV}$$

Example: Using formula of Binding energy, find the most stable isobar for $A = 25$.

Solution: $E_b = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} (\pm, 0) \frac{a_p}{A^{3/4}}$

$$\frac{dE_b}{dZ} = 0 \Rightarrow \frac{a_c}{A^{1/3}} (2Z-1) + \frac{4a_A}{A} (A-2Z) = 0$$

$$Z = \frac{a_c A^{-1/3} + 4a_A}{2a_c A^{-1/3} + 8a_A A^{-1}} = \frac{0.6 A^{-1/3} + 76}{1.19 A^{-1/3} + 152 A^{-1}}$$

For $A = 25$, $Z = 11.7 \approx 12$

For $A = 25$ there are three nuclide ${}^{25}_{11}\text{Na}$, ${}^{25}_{12}\text{Mg}$ and ${}^{25}_{13}\text{Al}$

So most stable is ${}^{25}_{12}\text{Mg}$.

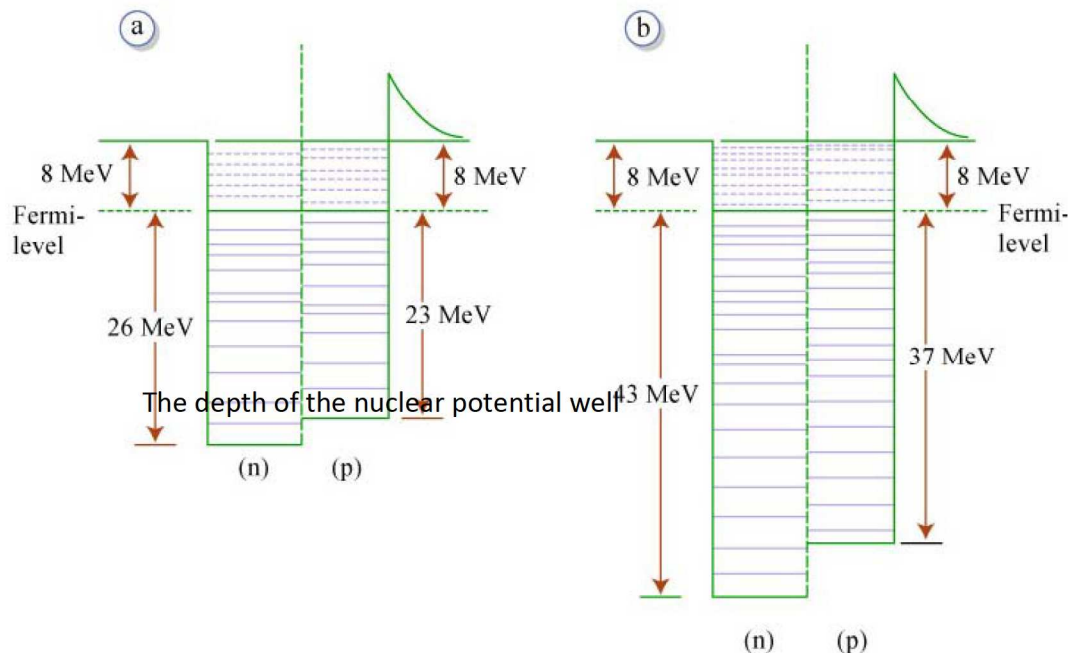
The depth of the nuclear potential well

From the Fermi energy we know (approximately) the energy difference between the bottom of the potential well and the energy of the highest occupied state.

We also know from the binding energies that the last nucleon is bound by approximately 8 MeV

From that we can conclude that the depth of the nuclear potential well is approximately $33 + 8 = 41$

We get the same depth for protons and neutron since we assumed the equal number of both. If we take into account the fact that the difference in the number of protons and neutrons we will get different depths of the potential wells for each species.



The depth difference for protons and neutrons

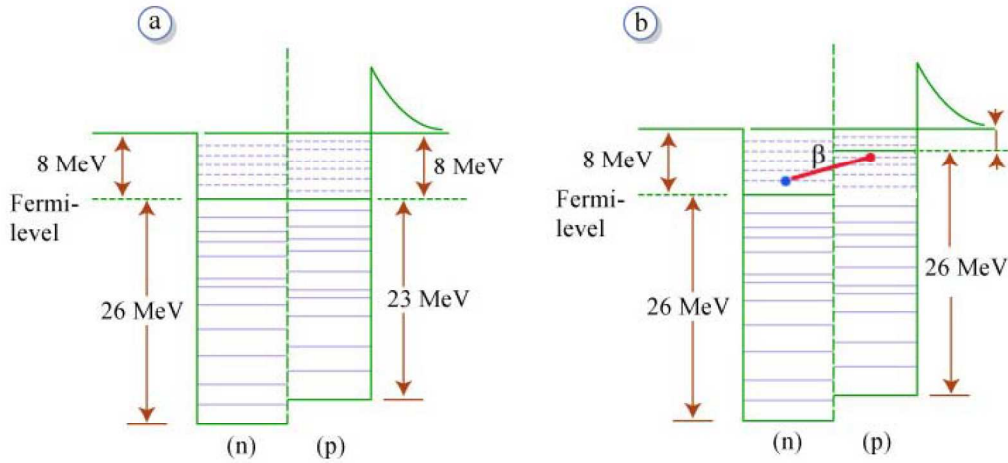
- (1) The potential depth difference comes out from the dependence of the Fermi energy on the number of protons or neutrons.
- (2) The difference in depth is a consequence of the Coulomb repulsion between the protons.

Beta decay and fermi energy

In stable nuclei the Fermi level is at the same energy for protons in neutrons. In such a case there is no energy gain from transforming one type of nucleons into another by β -decay.

In unstable nuclei the Fermi level is different for protons and neutrons, this opens a path to transform nucleons from one well to the other through β -decay. The β decay proceeds until Fermi levels in both wells are equal.

Fermi level for protons in neutrons in stable nuclei (a) and in unstable nuclei (b).



Average momentum of a nucleon

The average momentum of a nucleon depends on the density of momentum states which is the number of momentum states dn for a particle with momentum p and $p + dp$.

We can calculate this number using a similar trick to what we used to calculate the number of states up to the Fermi momentum.

Let us consider the space defined by the quantum numbers (n_x, n_y, n_z) . We argued there are two momentum states per unit volume in this space.

Particles with momentum between p and $p + dp$ define a spherical shell with radius $R = pL / \pi \hbar$ and thickness $dR = dpL / \pi \hbar$

in this space with volume $4\pi R^2 dR = 4\pi \left(\frac{L}{\pi \hbar} \right)^3 p^2 dp = \frac{4V}{\pi^2 \hbar^3} p^2 dp$

To get the number of states we need to remember that the condition for $n_x > 0, n_y > 0$ and $n_z > 0$ requires that we take $1/8$ of the volume V , and that we also need to multiply V by 2 to account for two spins state of the nucleon.

The density of states is

$$dn = 2 \frac{1}{8} V = \frac{1}{4} \frac{4V}{\pi^2 \hbar^3} p^2 dp = \frac{V}{\pi^2 \hbar^3} p^2 dp \Rightarrow \frac{dn}{dp} = \frac{V}{\pi^2 \hbar^3} p^2$$

The average momentum of a nucleon is

$$p_{av} = \frac{\int_0^{p_F} p \frac{dn}{dp} p^2 dp}{\int_0^{p_F} \frac{dn}{dp} p^2 dp} = \frac{\int_0^{p_F} p^3 dp}{\int_0^{p_F} p^2 dp} = \frac{3}{4} p_F \approx 188 \text{ MeV}$$

The average energy of a nucleon

To calculate the average energy, we need to know the density of energy state, which is the number of states for a particle with energy between E and dE .

We can get this number from the density of the momentum states

$$E = \frac{p^2}{2m} \Rightarrow p^2 = 2mE \Rightarrow p dp = m dE \text{ and } dp = \frac{m dE}{\sqrt{2mE}} = \sqrt{\frac{m}{2}} \frac{dE}{\sqrt{E}}$$

$$dn = \frac{V}{\pi^2 \hbar^3} p^2 dp \quad dn = \sqrt{2} \frac{V}{\pi^2 \hbar^3} m^{3/2} \sqrt{E} dE$$

The average energy of a nucleon

The average energy of a nucleon is

$$E_{av} = \frac{\int_0^{E_F} E \frac{dn}{dE} dE}{\int_0^{E_F} \frac{dn}{dE} dE} = \frac{\int_0^{E_F} E^{\frac{3}{2}} dE}{\int_0^{E_F} E^{\frac{1}{2}} dE} = \frac{3}{5} E_F \approx 20 \text{ MeV} \quad (25)$$

The energy which corresponds to the average momentum is

$$E' = \frac{p_{av}^2}{2m} = \frac{9}{16} \frac{p_F^2}{2m} = \frac{9}{16} E_F \approx 18.6 \text{ MeV} \quad (26)$$

Total energy in a nucleus

Knowing the average energy of a nucleon we can calculate the total energy of a nucleus

$$E_{tot} = N \langle E(p) \rangle + Z \langle E(n) \rangle = N \frac{3}{5} E_F(p) + Z \frac{3}{5} E_F(n)$$

Assuming for simplicity equal mass for a proton and a neutron

$$E_{tot} = \frac{3}{10} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \frac{\hbar^2 c^2}{mc^2 R_0^2} \frac{N^{\frac{5}{3}} + Z^{\frac{5}{3}}}{A^{\frac{2}{3}}}$$

Let us investigate the term which depends on N, Z and A further as it leads to some interesting consequences.

The term dependent on N, Z and A expands to

$$\frac{N^{\frac{5}{3}} + Z^{\frac{5}{3}}}{A^{\frac{2}{3}}} = A \left(1 + \frac{5}{9} \left(\frac{N-Z}{A} \right)^2 \right)$$

Therefore, the total energy has two terms

$$E_{tot} = \frac{3}{10} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \frac{\hbar^2 c^2}{mc^2 R_0^2} A + \frac{1}{6} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \frac{\hbar^2 c^2}{mc^2 R_0^2} \left(\frac{N-Z}{A} \right)^2$$

The first term corresponds to the volume term in the liquid drop model while the last one corresponds to the symmetry energy term in the liquid drop model.

Therefore, based on the Fermi model we can conclude that the symmetry energy term in the liquid drop model is a quantum mechanical effect related to the way fermion occupy allowed states in the proton/neutron potential well.

1.4 Stable Nuclei

Not all combination of neutrons and protons form stable nuclei. In general, light nuclei ($A < 20$) contain equal numbers of neutrons and protons, while in heavier nuclei the proportion of neutrons becomes progressively greater. This is evident in figure as shown below, which is plot of N versus Z for stable nuclides.

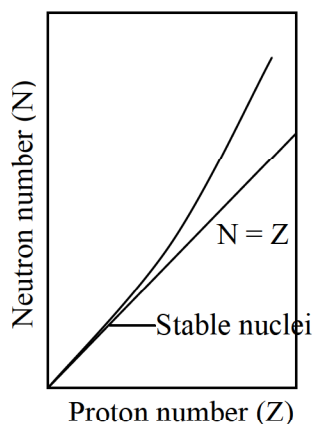


Figure: Neutron-proton diagram for stable nuclei

The tendency for N to equal Z follows from the existence of nuclear energy levels. Nucleons, which have spin $\frac{1}{2}$, obey exclusion principle. Hence proton and neutrons are fermions they follow Pauli exclusion principle. As a result, each energy level can contain two neutrons of opposite spins and two protons of opposite spins. Energy levels in nuclei are filled in sequence, just as energy levels in atoms are, to achieve configurations of minimum energy and therefore maximum stability.

Thus the boron isotope $^{12}_5\text{B}$ has more energy than the carbon isotope $^{12}_6\text{C}$ because one of its neutrons is in a higher energy level, and $^{12}_5\text{B}$ is accordingly unstable. If created in a nuclear reaction, a $^{12}_5\text{B}$ nucleus changes by beta decay into a stable $^{12}_6\text{C}$ nucleus in a fraction of second.

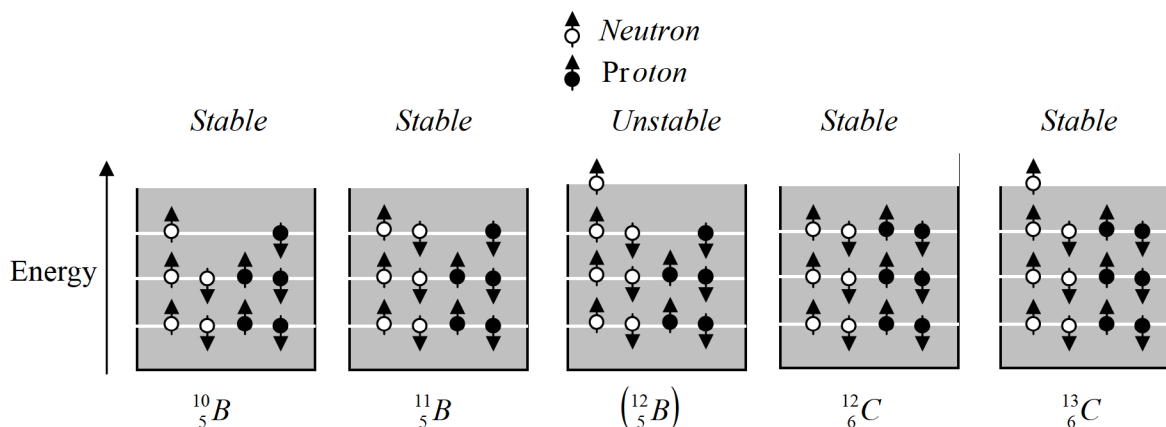


Figure: Simplified energy level diagrams of some boron and carbon isotopes.

The preceding argument is only part of the story. Protons are positively charged and repel one another electrically. This repulsion becomes so great in nuclei with more than 10 protons or so that an excess of neutrons, which produce only attractive forces is required for stability. Thus the curve departs more and more from $N = Z$ line as Z increases.

Sixty percent of stable nuclides have both even Z and N ; these are called “**even-even**” nuclides. Nearly all the others have either even Z and odd N (“**even-odd**” nuclides) or odd Z and even N (“**odd-even**” nuclides) with the numbers of both kinds being about equal.

Only five stable **odd-odd** nuclides are known: ^2_1H , ^6_3Li , $^{10}_5\text{B}$, $^{14}_7\text{N}$ and $^{180}_{73}\text{Ta}$. Nuclear abundances follow a similar pattern of favoring even numbers for Z and N .

These observations are consistent with the presence of nuclear energy levels that can each contain two particles of opposite spin. Nuclei with filled levels have less tendency to pick up other nucleons than those with partially filled levels and hence were less likely to participate in the nuclear reactions involved in the formation of elements.

Nuclear forces are limited in range, and as a result nucleons interact strongly only with their nearest neighbors. This effect is referred to as the **saturation** of nuclear forces. Because the coulomb repulsion of protons is appreciable throughout the entire nucleus, there is a limit to the ability of neutrons to prevent the disruption of large nucleus. This limit is represented by the bismuth isotope $^{209}_{83}\text{Bi}$, which is the **heaviest stable** nuclide.